



EECS 570 Programming Assignment 1

August 29th, 2025

About Me

- Education
 - First-semester SUGS student in ECE
 - Graduated this past May from Michigan in CSE with EE minor
- Teaching
 - EECS 482 IA
 - EECS 570 IA
 - This class!
- Outside the classroom
 - Soccer
 - Snowboarding

Announcements

- Final Project
 - Piazza - search for teammates
 - Next Friday: Project introduction / handout
- PA1
 - Due 9/24 on Canvas
 - Please pay attention to submission format!
- Office Hours
 - Tue & Thu: 2:00 - 3:30 PM, BBB Atrium

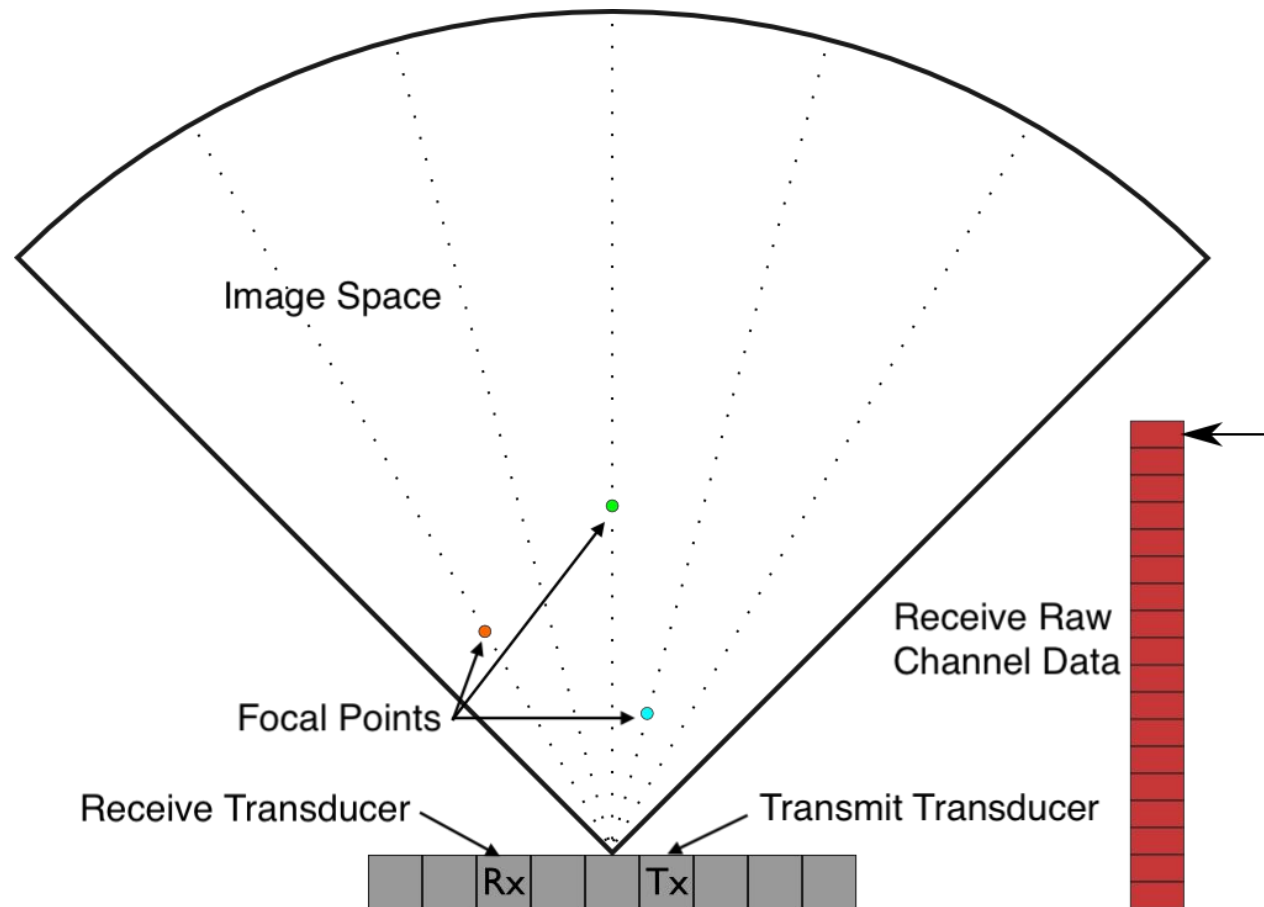
Overview

- Medical Imaging using Ultrasound
 - Introduction
 - Transmission and Reception
- Introduction to POSIX Threads
 - Thread Creation and Joining
 - Synchronization Primitives

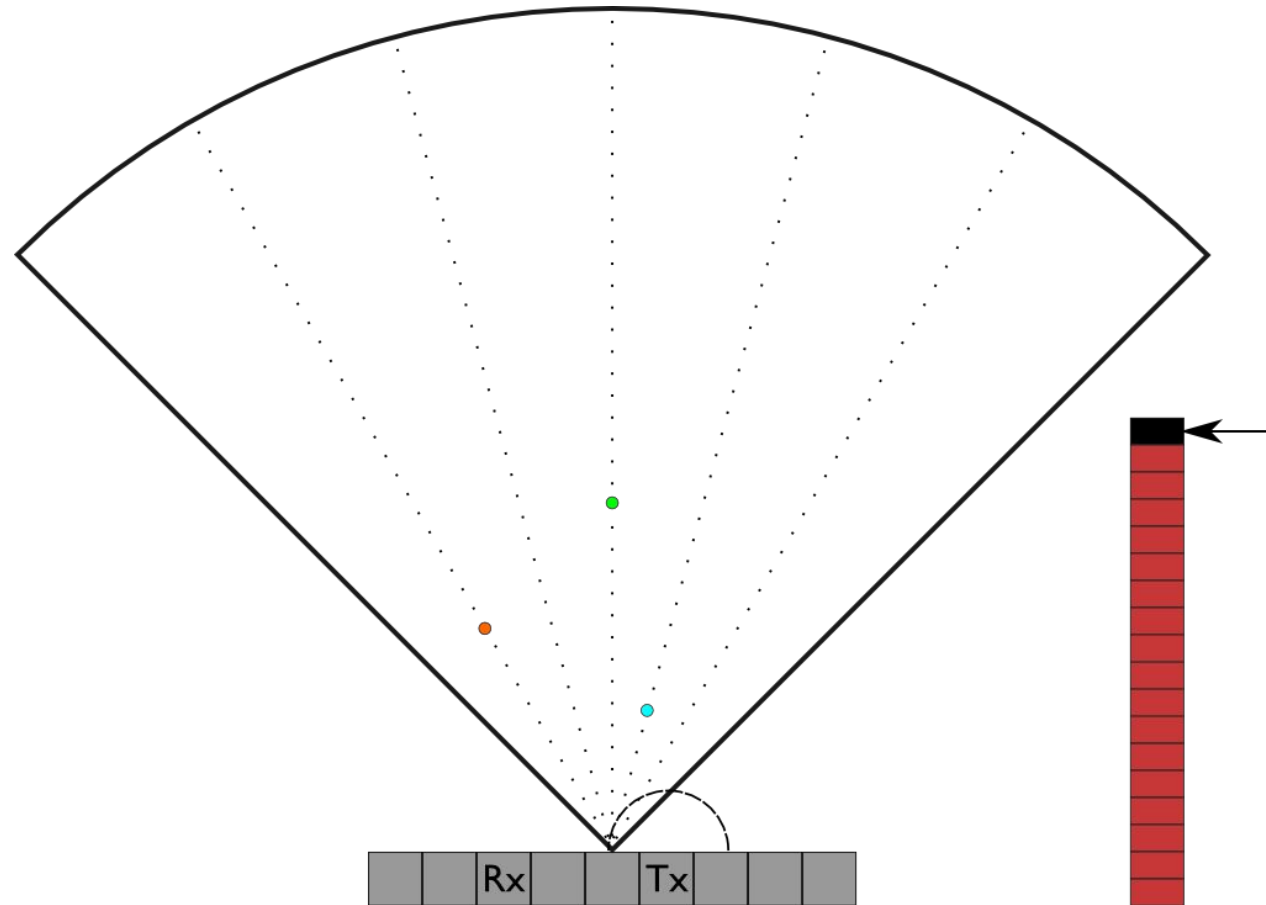
Portable Medical Imaging Devices

- Medical Imaging moving towards portability
 - MEDICS (X-Ray CT) [Dasika '10]
 - Handheld 2D Ultrasound [Fuller '09]
- Not just a matter of convenience
 - Improved patient health [Gunnarsson '00, Weinreb '08]
 - Access in developing countries
- Why ultrasound?
 - Low transmit power [Nelson '10]
 - No danger or side-effects

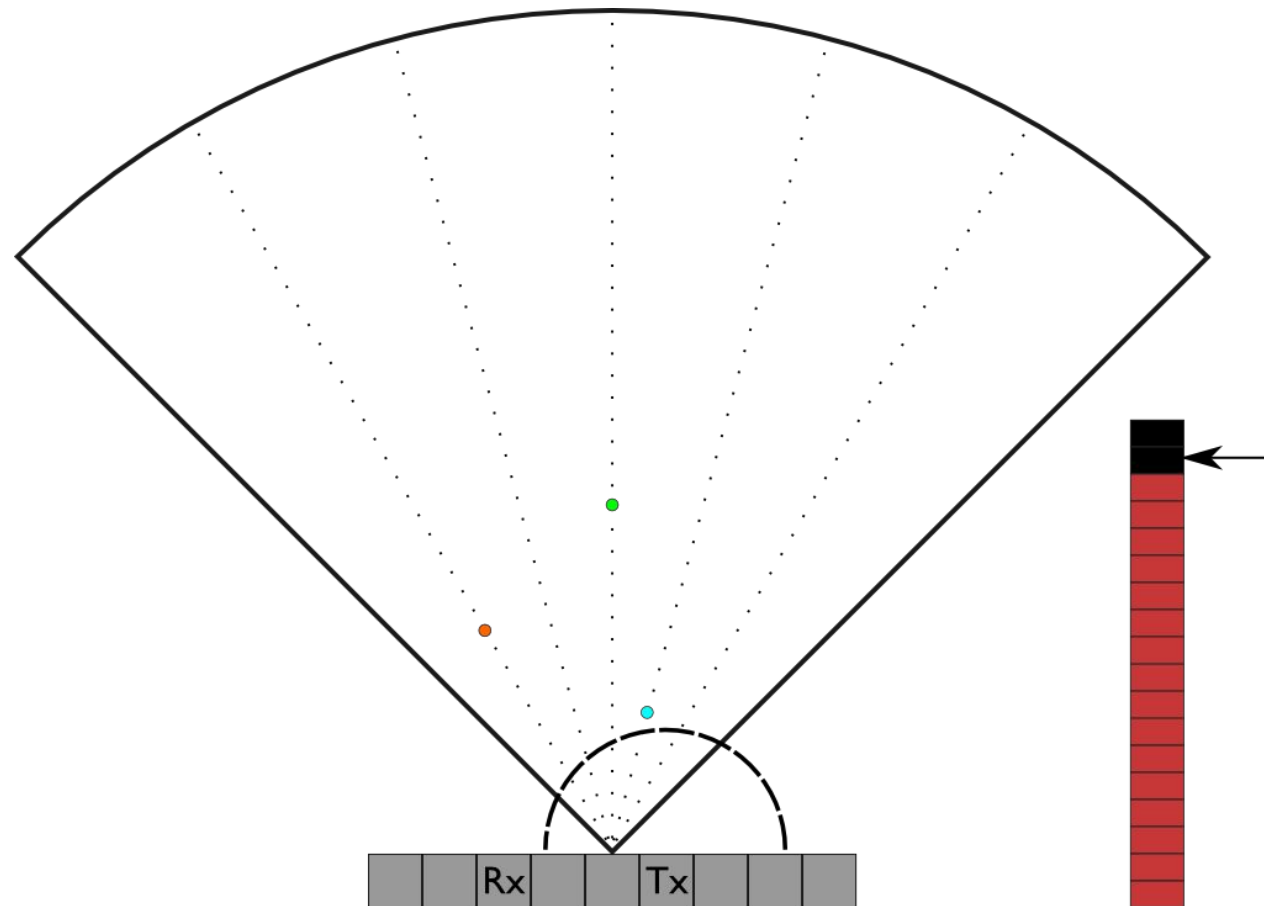
Ultrasound: Transmission and Reception



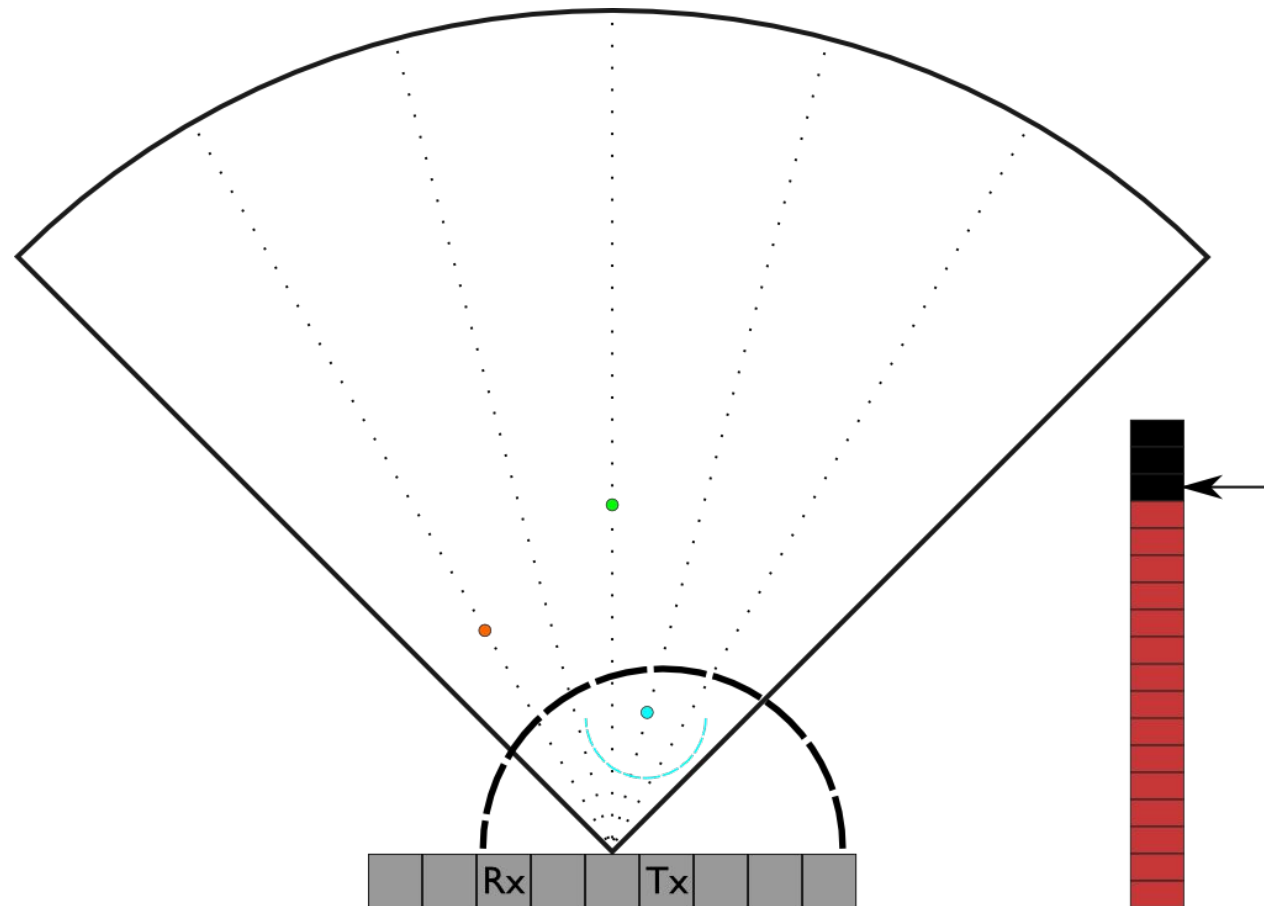
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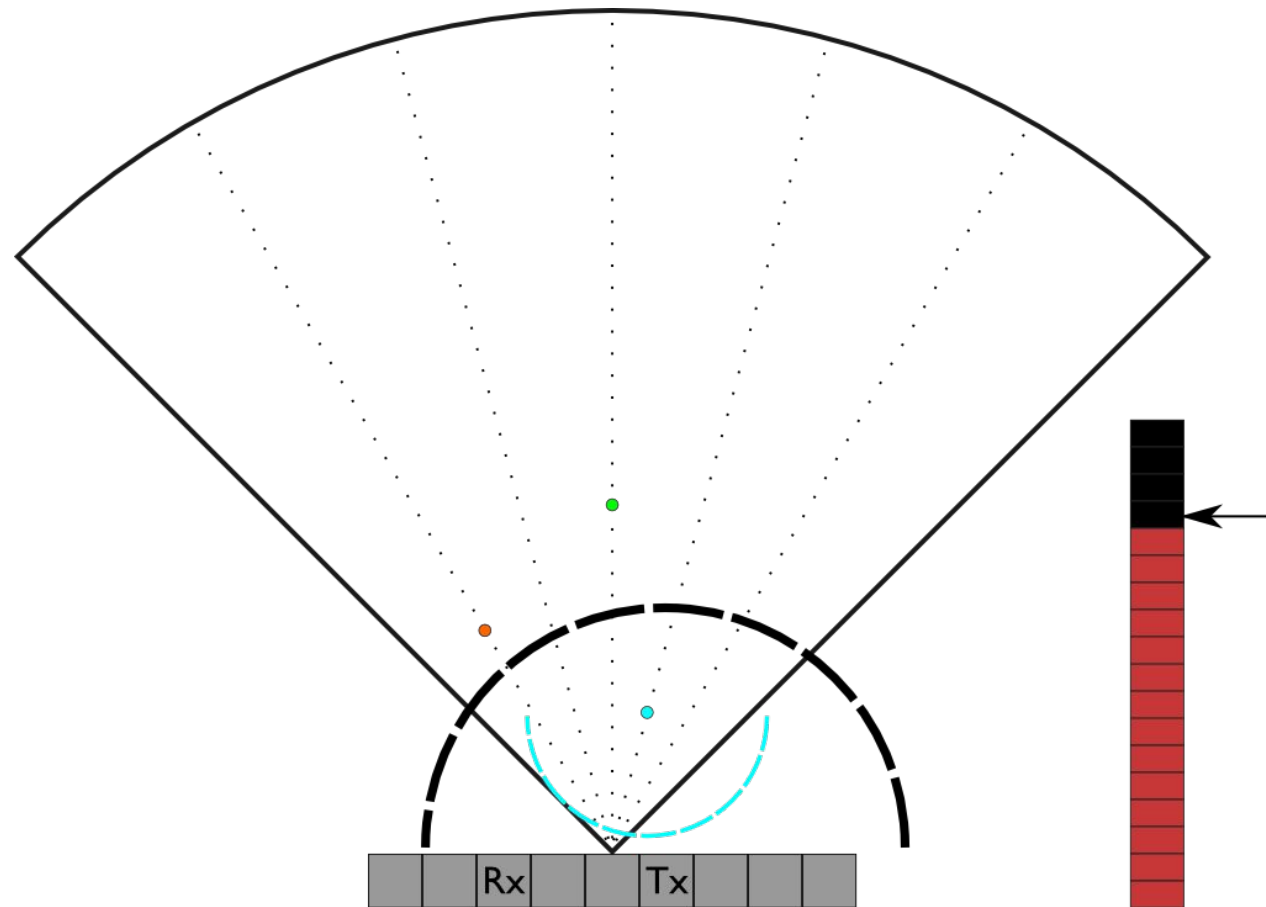
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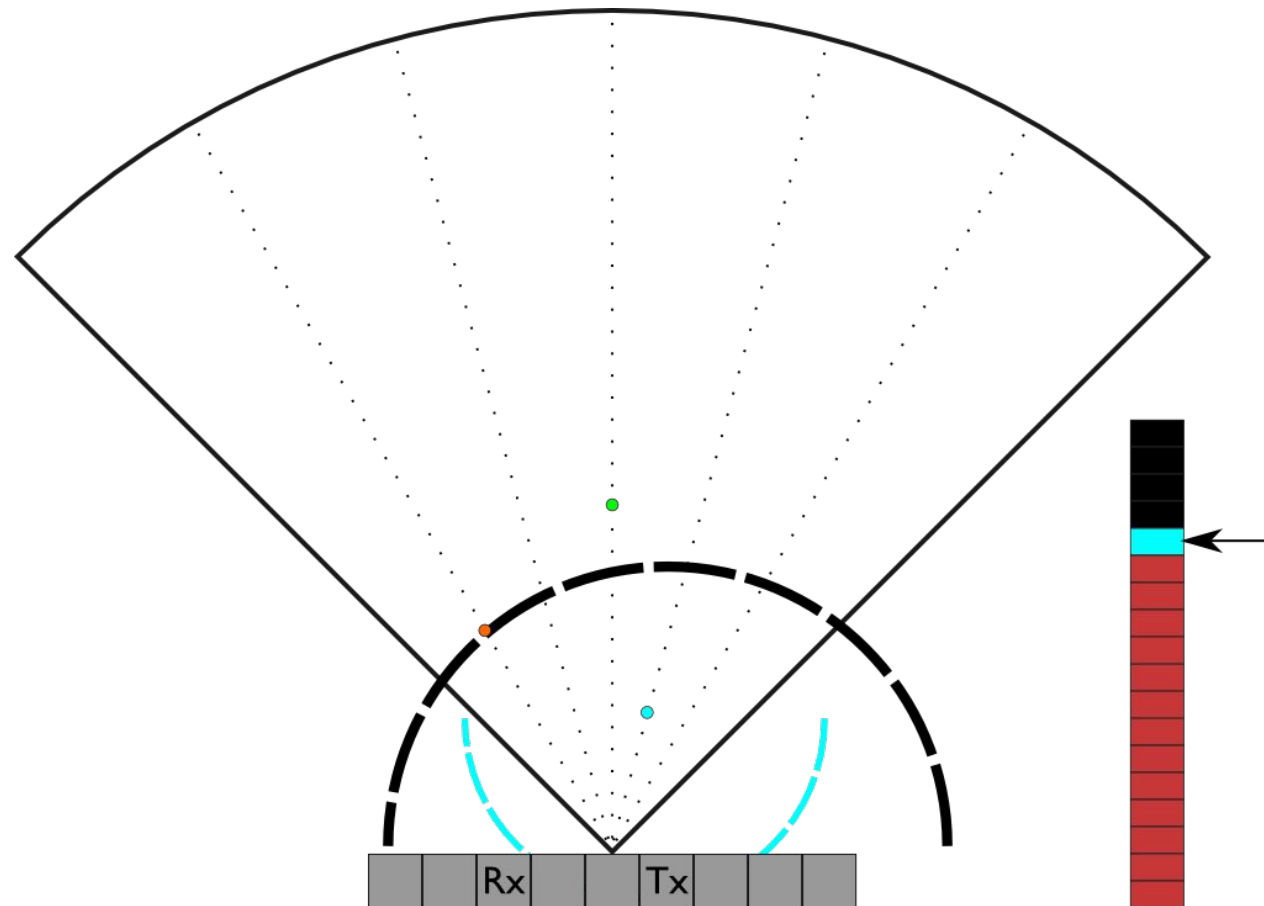
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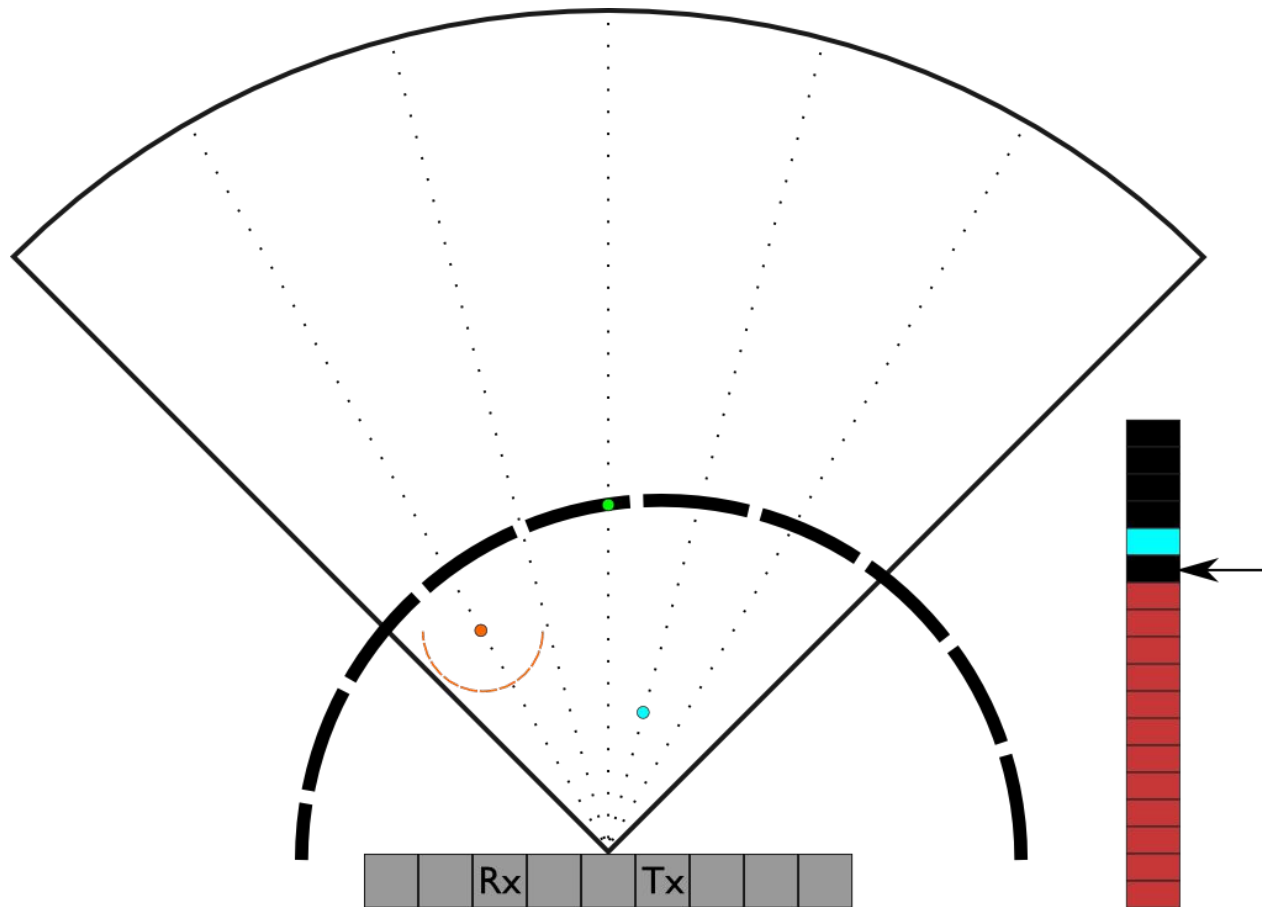
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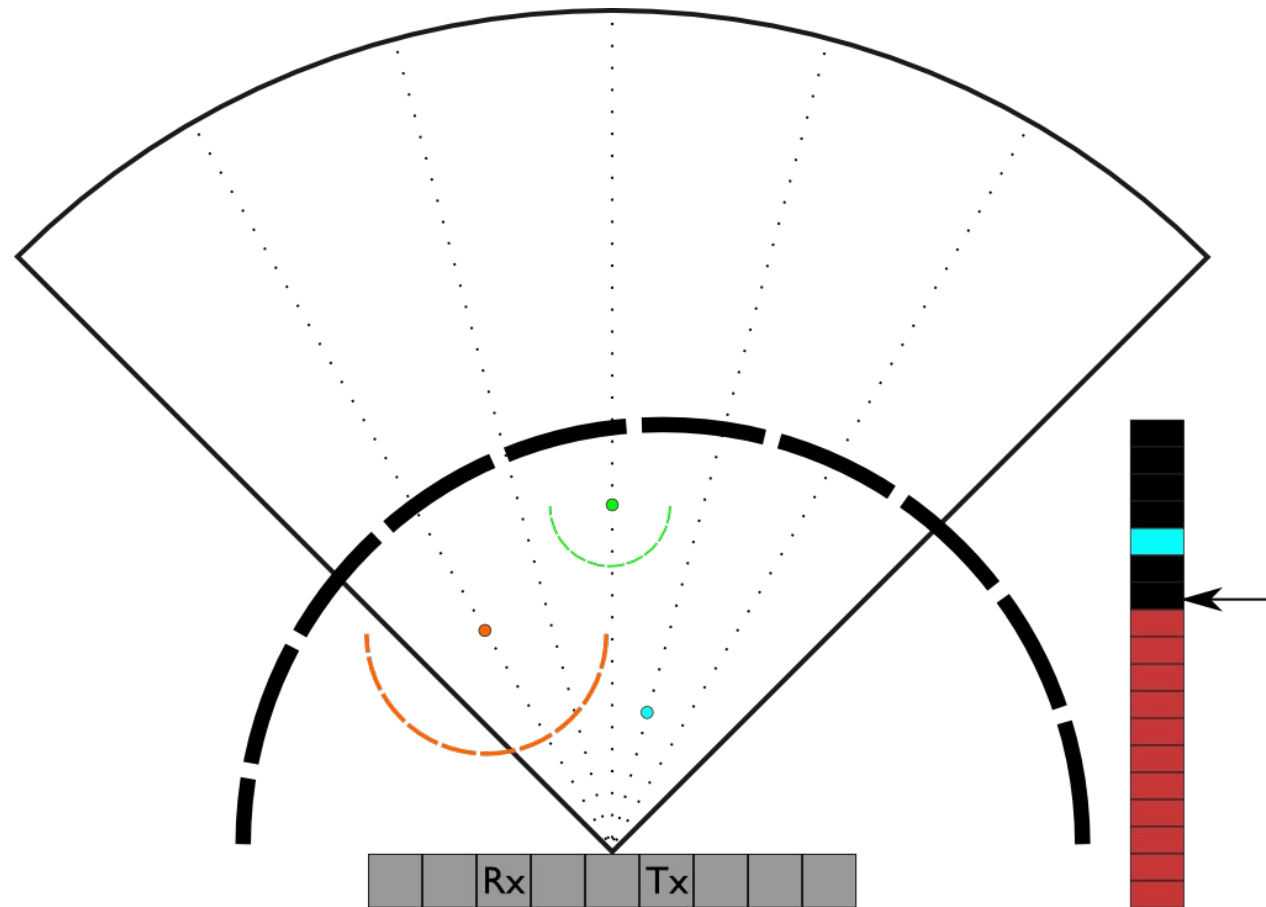
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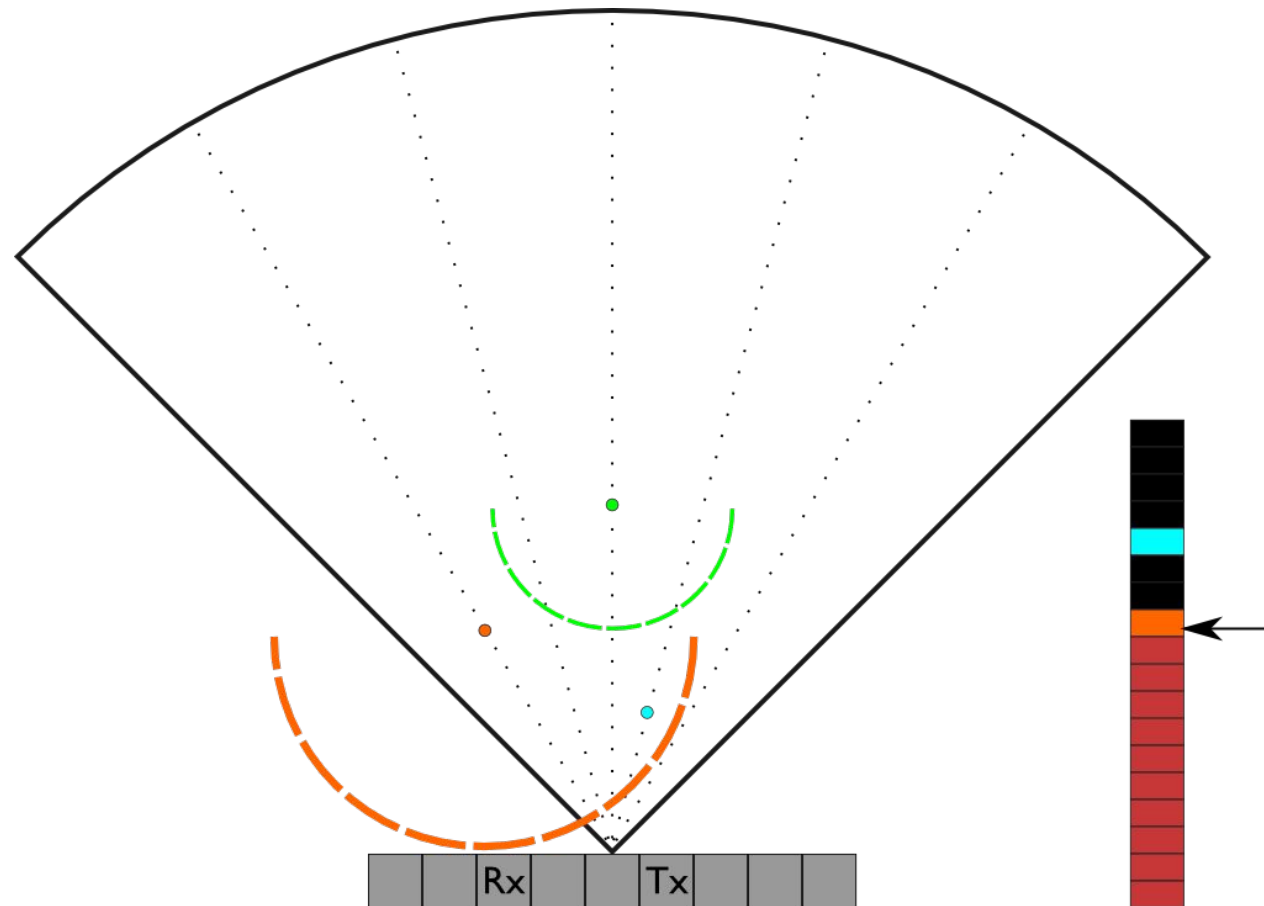
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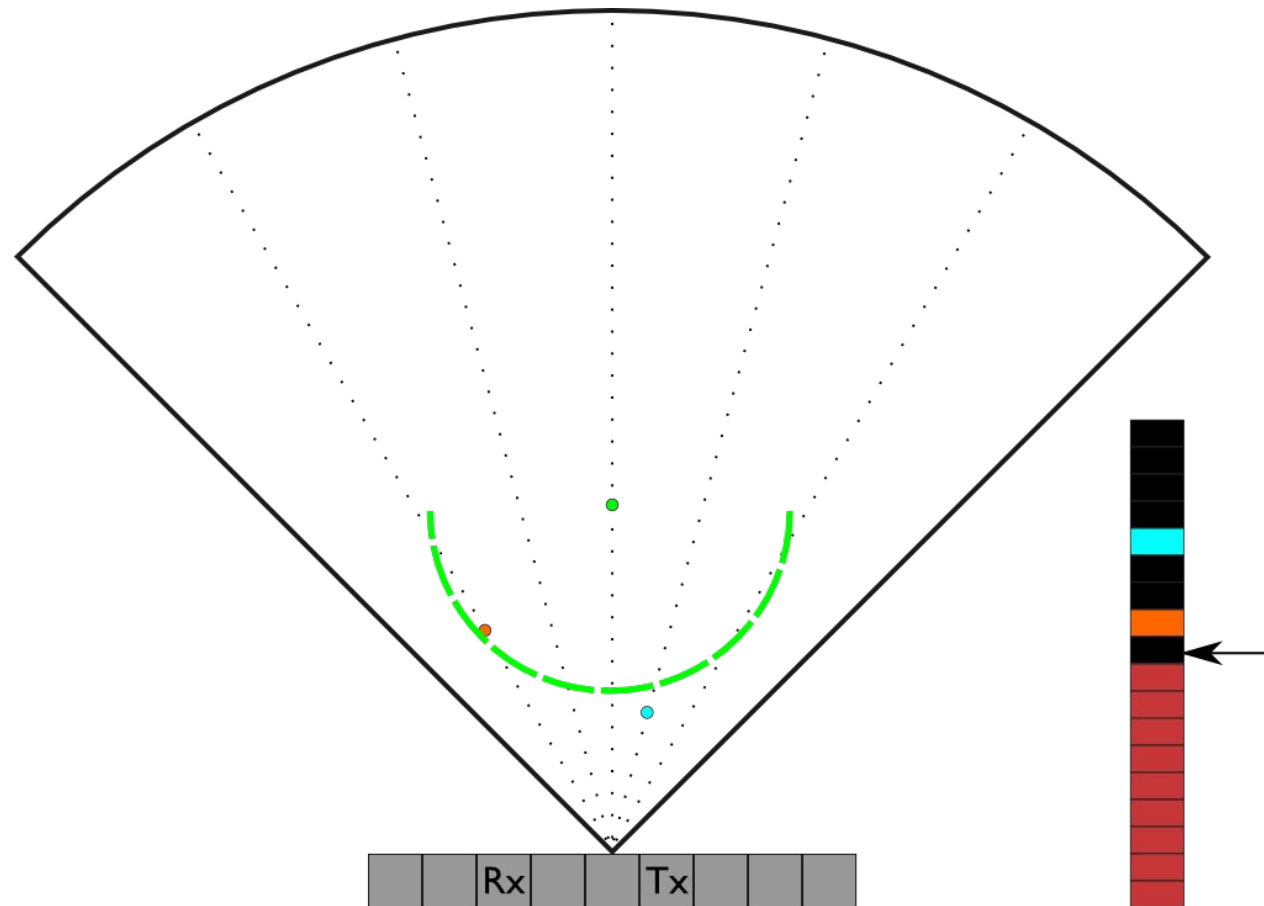
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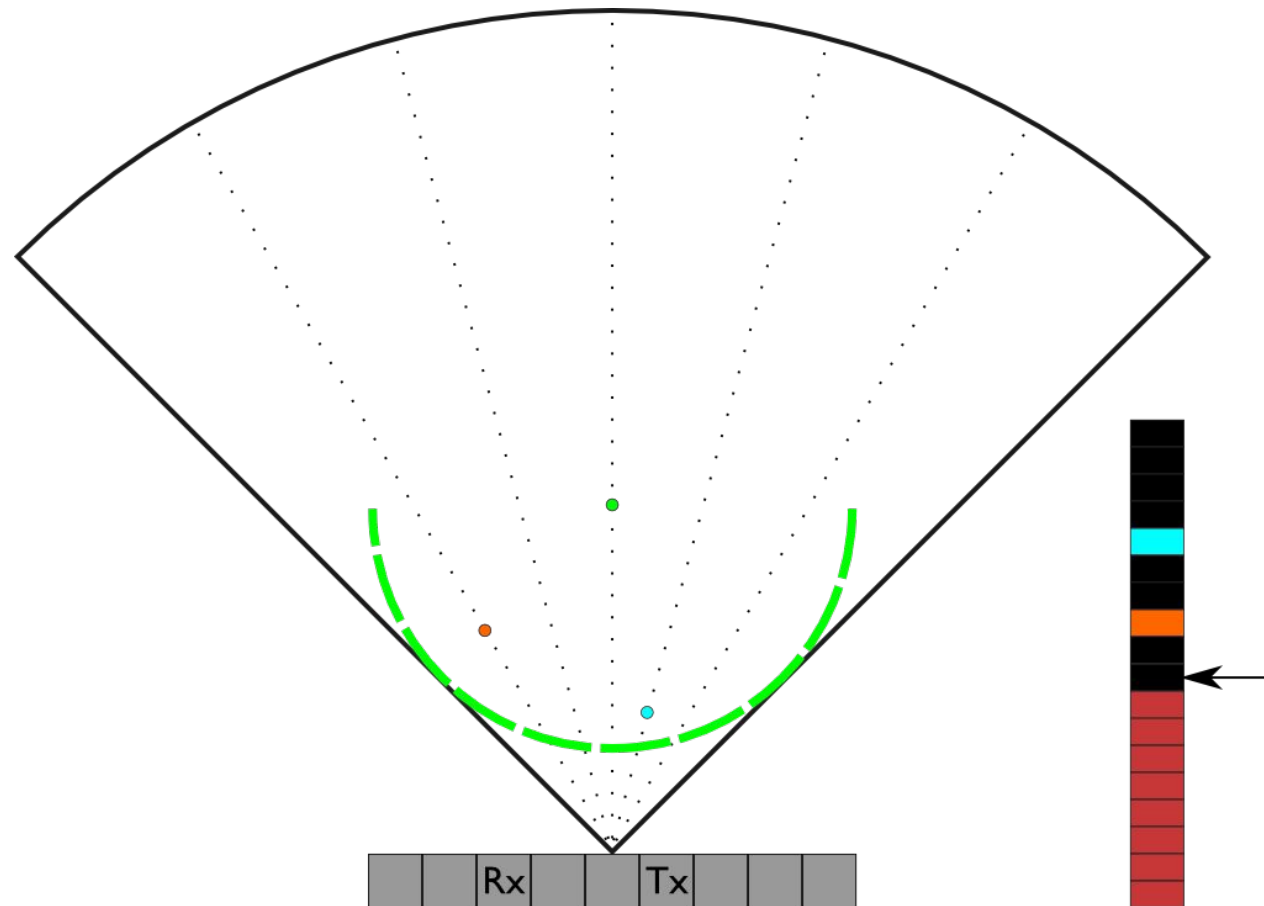
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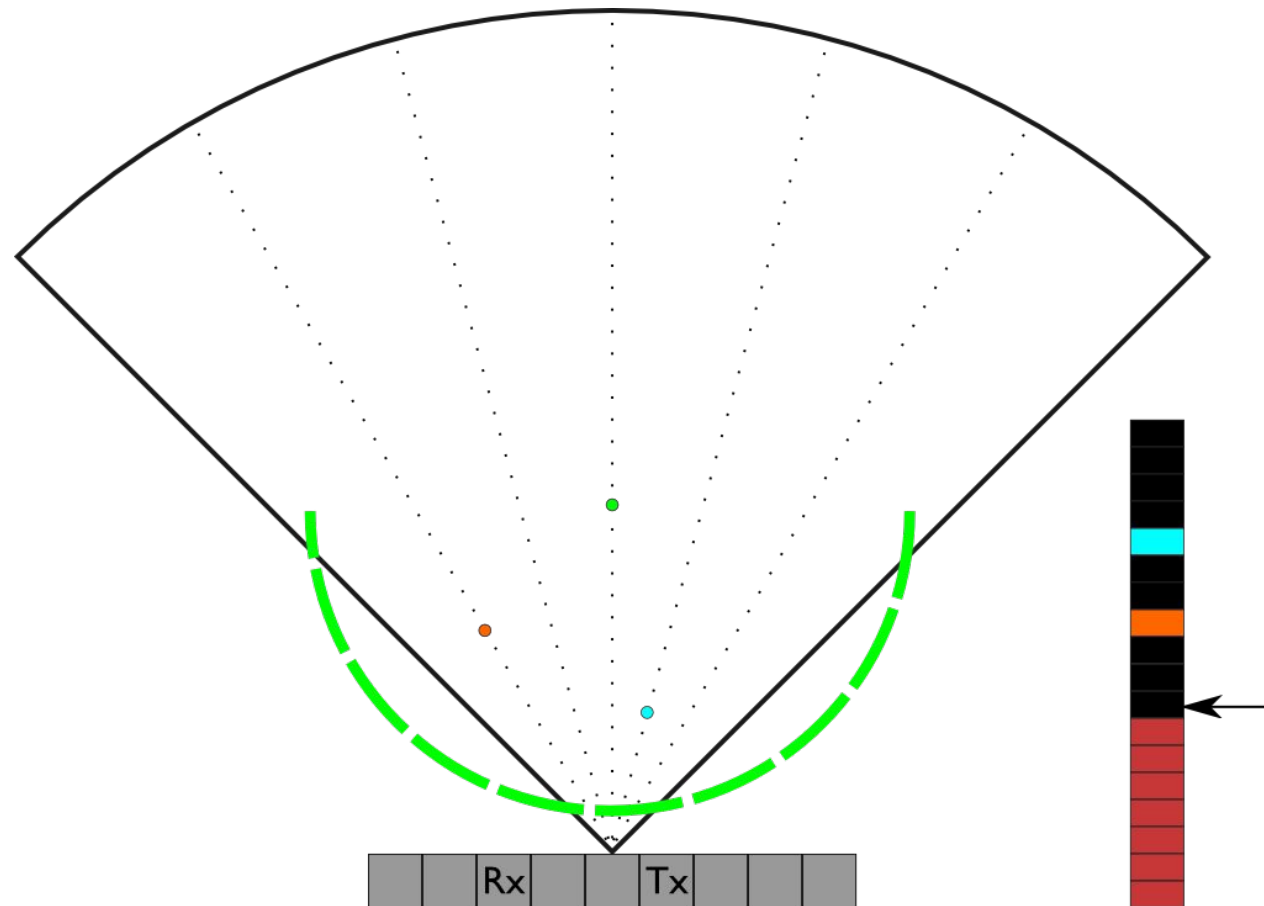
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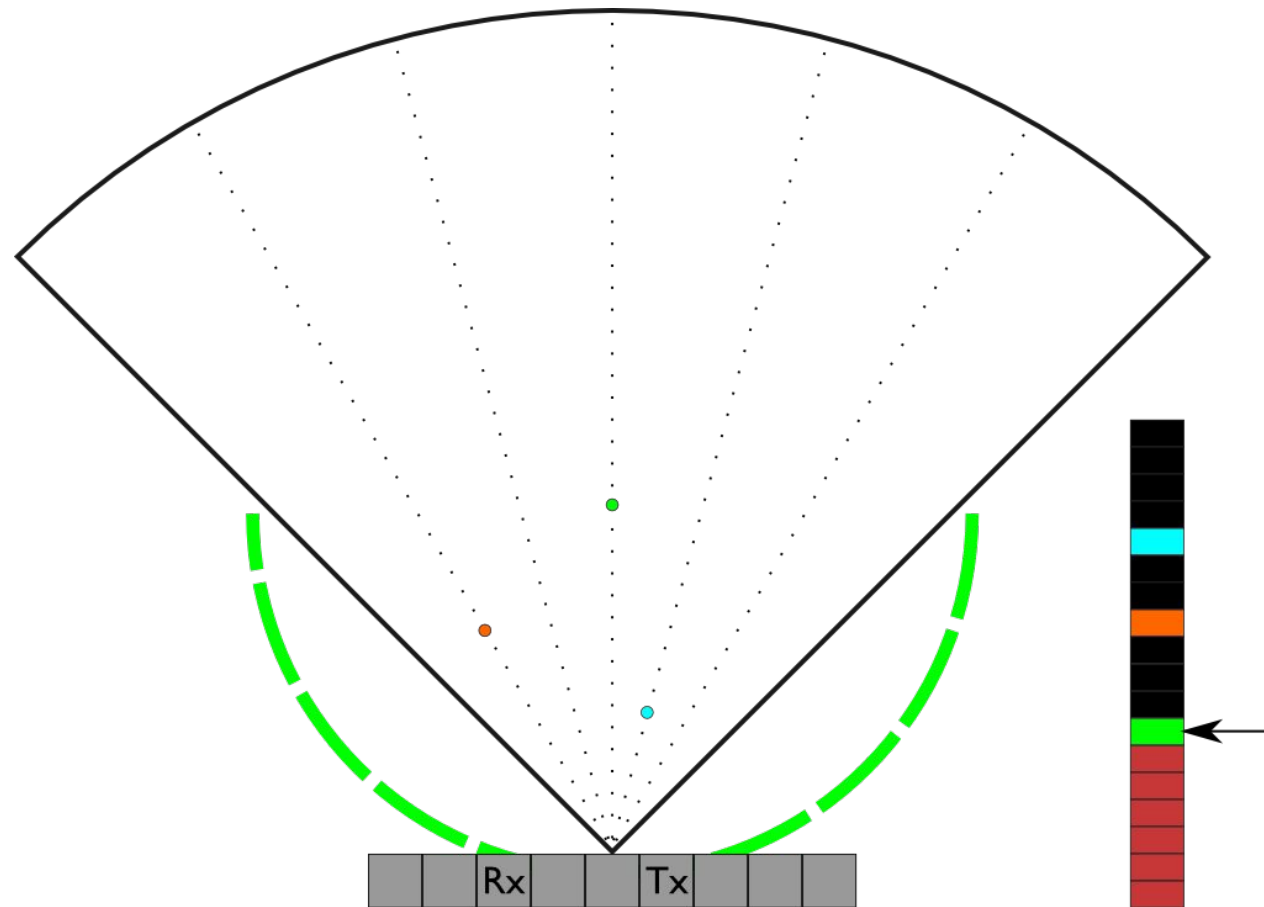
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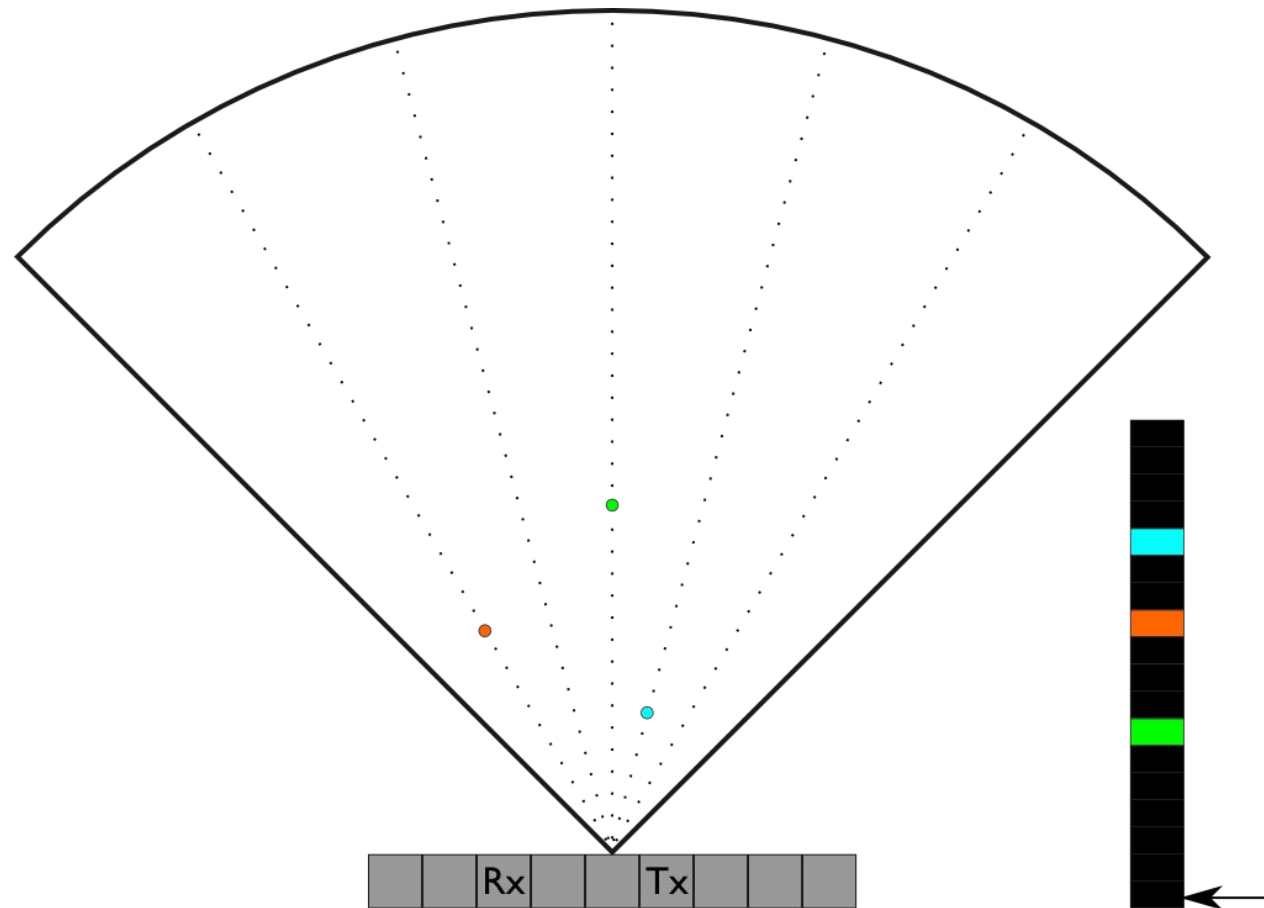
Ultrasound: Transmission and Reception



Ultrasound: Transmission and Reception



Ultrasound: Transmission and Reception



Each transducer stores an array of raw received data

Ultrasound: Transmission and Reception

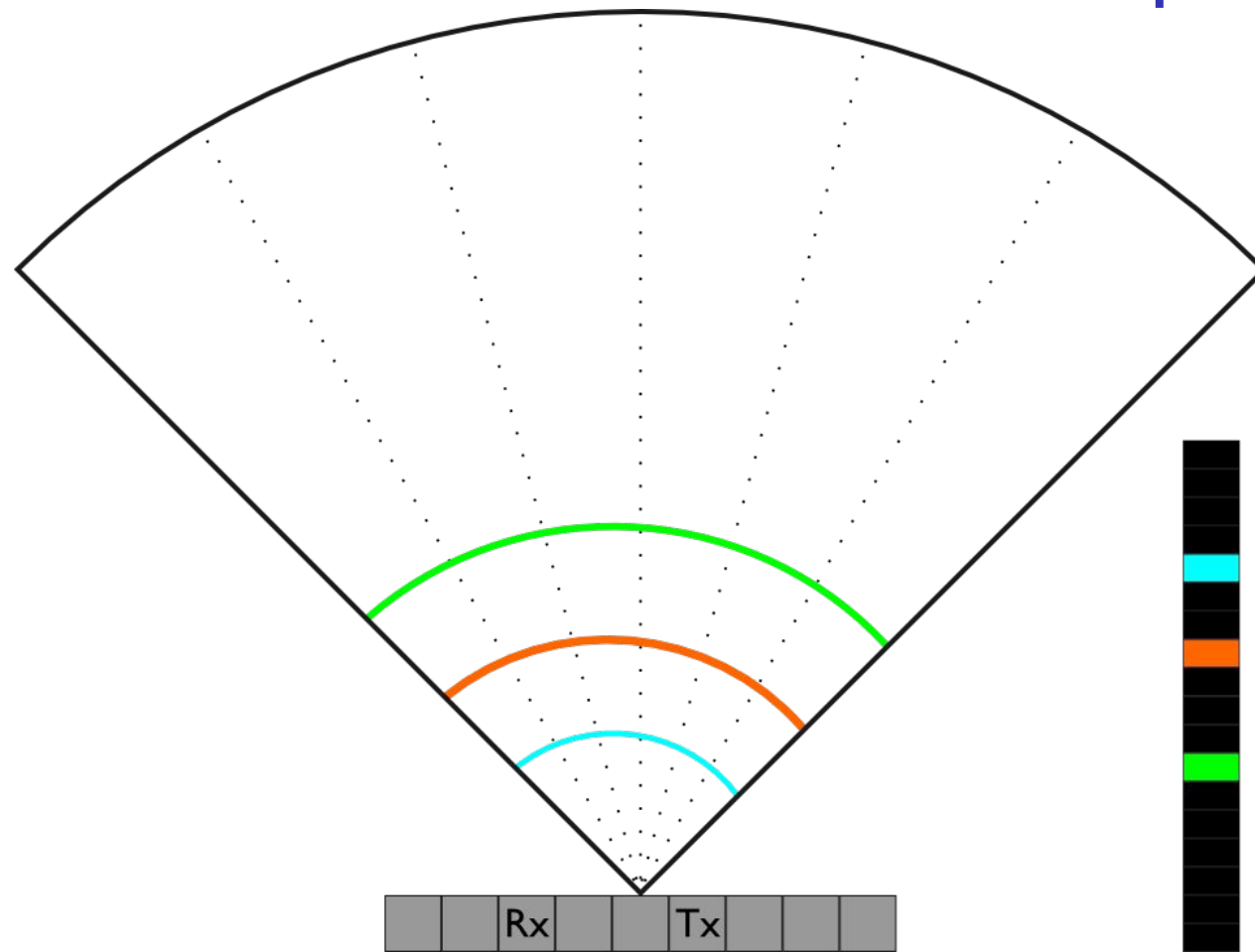
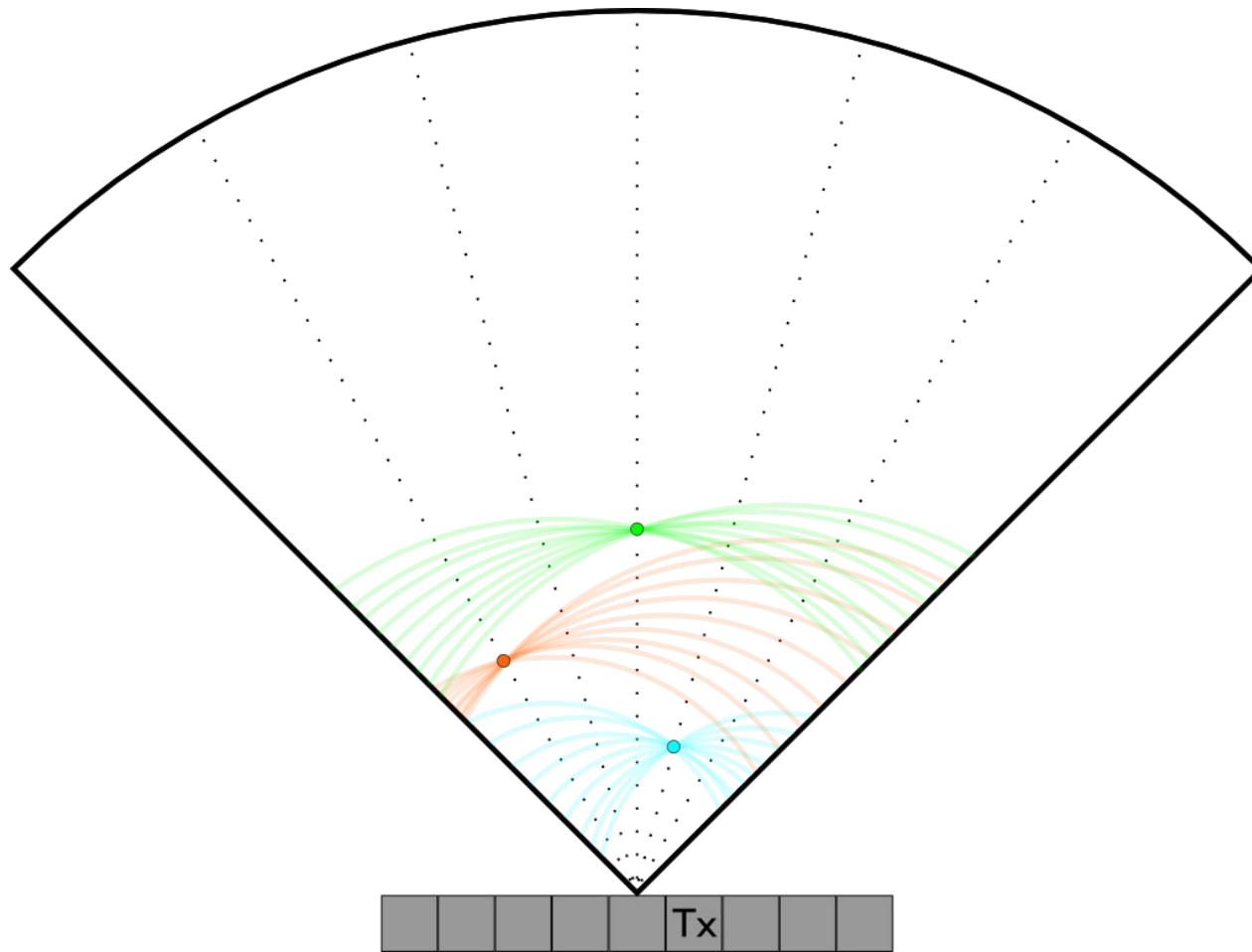


Image reconstructed from data based on round-trip delay

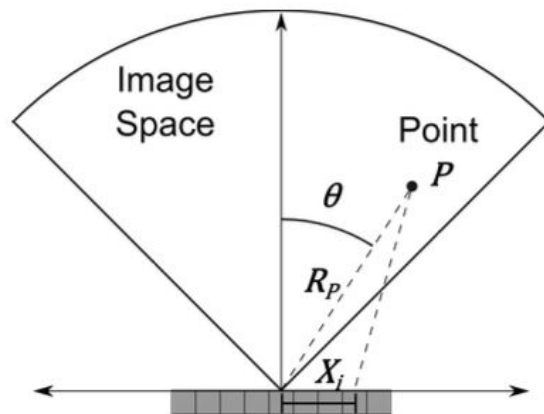
Ultrasound: Transmission and Reception



Images from each transducer combined to produce the full frame

Delay Index Calculation

Iterate through all index points for each transducer and calculate delay index τ_P



$$\tau_P = \frac{f_s}{c} (R_P + \sqrt{R_P^2 + X_i^2 - 2R_P X_i \sin \theta})$$

Often done with lookup tables (LUTs)
50 GB LUT required for target 3D system

Pthreads Tutorial

- What is a thread?

Pthreads Tutorial

- What is a thread?
 - Independently executable stream of instructions
 - Schedulable unit of execution for the operating system

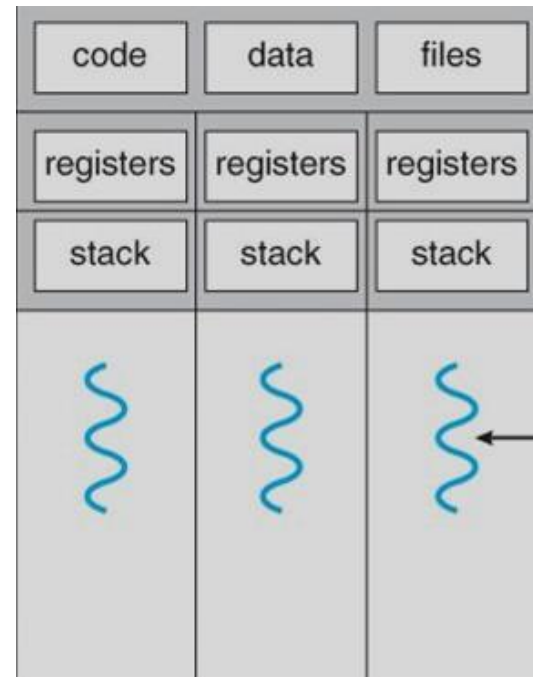
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Pthreads Tutorial

- What is a thread?
 - Independently executable stream of instructions
 - Schedulable unit of execution for the operating system
- Pthreads - the POSIX threading interface
 - Provides system calls to *create* and *synchronize* threads
 - Communication happens through shared memory
 - Specifically using *pointers* to shared data

Pthreads Tutorial



[Posix Threads Tutorial](#)